

K5 Expert Machine Operation and Maintenance Manual

Machine Serial Number: V18G019



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IMPORTANT INFORMATION

	Note!
	Read this manual, and any relevant manuals from third party vendors, before operating or maintaining this machine.

This manual was relevant to the machine at the time of writing. Due to continual improvements, final operating procedures may vary. We take care to ensure that information is correct, but BOBST will assume no responsibility for loss, damage, or injury for any inaccuracy. The company reserves the right to alter or modify this document at any time. If you have any questions, or require further information on your machine, please contact BOBST.

The system owner is responsible for ensuring that:

- Any person involved with starting up, operating, maintaining, and servicing the machine has read and understood the relevant sections of these operating instructions
- Users are technically qualified, appropriately trained and instructed, and have been informed of the potential hazards and risks
- Unauthorised persons are not able to access the machine
- Users comply with safety instructions and warnings in this document
- Users who load and unload substrates are adequately trained and informed about potential hazards.

Copyright Information

K5 Expert Edition

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PREFACE

This manual provides basic operation and maintenance knowledge for the machine. The manual is split into the following sections:

- [Section 1.0 Introduction & General Description on page 29](#)

This section describes the processes, and introduces sections and components of the machine.

- [Section 2.0 Vacuum Pumping System on page 40](#)

This section provides information on:

- vacuum pumping components
- control operational modes
- measurement systems used to monitor the vacuum levels
- types of pumps to evacuate the chamber to the required operating pressure level
- components used to evacuate the machine
- leak-detecting methods
- gauges used to monitor pressure levels, and their purpose.

- [Section 3.0 Aluminium Evaporation Process on page 77](#)

This section describes the coating process and relevant system components.

- [Section 4.0 Operation of the Machine Winding Process on page 144](#)

This section describes the coating process and relevant system components.

- [Section 5.0 Alubond Process on page 240](#)

This section describes the coating process and relevant system components.

- [Section 6.0 Plasma Treater Systems on page 285](#)

This section explains the purpose and usage of plasma treater components.

- [Section 7.0 Machine Operational Sequences on page 321](#)

This section describes the operation of the machine in a series of flow charts.

- [Section 8.0 Machine Cleaning Processes on page 348](#)

This section describes cleaning procedures in a normal production environment.

- [Section 9.0 Auxiliary equipment on page 369](#)

This section describes the auxiliary equipment on the machine.

- [Section 10.0 Process Consumables on page 504](#)

This section describes consumable items used in production.

- [Section 11.0 Machine Calibration Procedures on page 520](#)

This section describes the calibration procedures for the machine.

- [Section 12.0 Preventative Maintenance on page 558](#)

This section describes preventative maintenance tasks and lubrication guidelines.

- Appendices

This section provides supplemental information, data sheets, training manuals, and product updates as follows:

- **Appendix A - Bill of Materials:** Parts listing and drawing references
- **Appendix D - Class D Fire Extinguishers:** Information on the fire extinguisher recommended in this manual
- **Appendix E - International Standard Applied to Protective Clothing:** Description of the standard.

Safety Warnings

It is essential that this section of the manual is read and complied with to avoid injury or damage.

Manual Conventions

The following safety precautions have specific meanings in Bobst Manchester manuals:

	Note!
	Important information.

	Caution! Warning of machinery damage.
	Warning! Warning of personal injury, loss of life, or substantial machinery damage.
	DANGER! Risk of serious injury or death.
	DANGER! Risk associated with live electrical circuits.

Common Safety Symbols

The following international standard symbols are used within this manual also as signage on the machine warning labels:

				
Danger: electricity, electric shock (arc flash labels)	Attention notice	General warning	Hazardous materials, toxic substances	Magnetic fields that may influence the functioning of pacemakers
				
Harmful or irritant materials	Flammable materials	Hot surface hazard	Hazard warning sign: hand injury	Implosion hazard

				
Crushing hazard, danger of entrapment	Entrapment hazard	Danger: overhead crane or overhead load	Compressed gas bottle hazard	Potential hazard to pressurised system
				
Fork lift trucks operating	Hazardous to the environment	Beware of moving machinery	Explosion risk, explosive material, or compressed gases	Corrosive materials
				
Strong magnetic field hazard	Forklift trucks working in this area	Trip hazard	Low temperature hazard	Cutting hazard, sharp edges, risk of hand injury
				
Magnetic fields influence	Slippery floor, slippery surface	No smoking	Oxidiser	Prohibited
				
Cut hazard	Asphyxiation hazard			

Personnel Protection Equipment Advice Symbols

The following international standard symbols are used within this manual and also as signage on the machine warning labels:

 Wear protective clothing	 Wear gloves	 Wear goggles	 Wear mask
Wear protective clothing	Wear gloves	Wear goggles	Wear suitable mask

Machine Identification Serial Number

The serial number is located on the equipment identification label on the rear side of the chamber, near the viewing window.

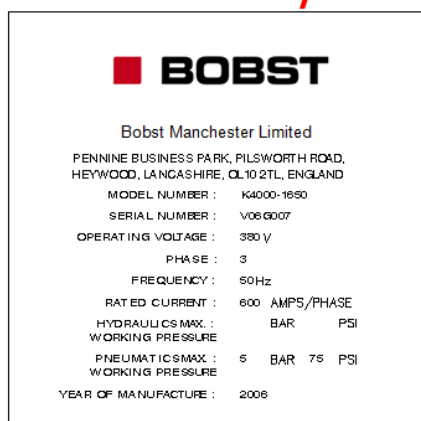
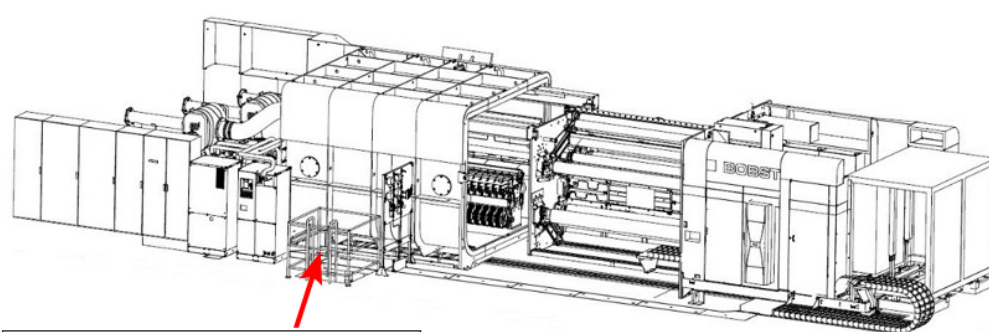


Figure 1.4-1 Sketch showing location of equipment serial plate

The equipment serial plate includes the following information

- **Model Number:** model type
- **Serial Number:** unique reference number for the machine
- **Operating Voltage**
- **Phase:** power supply type
- **Frequency:** power supply frequency
- **Rated Current:** full load installed current per phase
- **Hydraulic Max. Working Pressure:** rated maximum working pressure in Bar and PSI
- **Pneumatic Max. Working Pressure:** rated maximum working pressure in Bar and PSI
- **Year of Manufacture.**

Please include the serial number when contacting Bobst Manchester.

Customer Support and Aftersales Information

Bobst Manchester strives to provide fast, courteous customer support for:

- Safety information
- Warranty enquiries
- Technical support
- Machine upgrades
- Spares parts
- Service requests
- Site visits.

If you need assistance, please email services.manchester@bobst.com or write to Bobst Manchester Ltd., Pennine Business Park, Heywood, Lancashire, OL10 2TL, UK.

Telephone (+44) 1706 622 442, or fax (+44) 1706 622 772.

Technical Contact Numbering (TCN)

A TCN reference number is assigned to customer support requests. Use this to identify ongoing support tickets with BOBST.

MACHINE SAFETY

This machine has been manufactured to meet, or exceed, all codes and industry standards for machine safety.

All safety guidelines must be adhered to during the processes described in this manual and any other named process. All other uses are improper and can result in injury or equipment damage. BOBST will assume no liability for any such injuries or damage.

All personnel operating or maintaining the machine must follow these general safety guidelines:

- Always wear safety shoes
- Do not wear neckties, long sleeves, wristwatches, rings, gloves, etc.
- Keep the area around the machine free from obstructions
- Keep the area around the machine dry
- Follow your company's guidance on lifting heavy objects
- Make sure that access platforms are sturdy and have non-slip surfaces
- Keep tools and parts off the machine
- Do not use the machine as a work surface
- Use approved eye protection and air filtration equipment when cleaning the machine.

Read this section in conjunction with the safety instructions in the rest of this manual.

Correct Use of Equipment

	Note!
	The process materials listed below are specified for operation of this machine. The vacuum pumping system is designed for the specified gases. Use of other process materials or gases is improper.

The machine is designed to coat one side of a flexible material, or substrate. Coatings are achieved using aluminium, with or without inline de-metalization.

Specified Process Materials

Process materials	Designation
Pure Gases or a combination	Argon Oxygen
Evaporating materials	Aluminium
Inline Plasma System Target material (specification related)	Aluminium Stainless Steel

Table 1.5-1 Specified Process Materials

	Warning!
	No other materials than those mentioned above should be evaporated in this machine.

Coating Processes

This manual provides guidance on aluminium, the standard coating that is to be carried out on this machine. Specific safety considerations apply to other processes.

Aluminium reacts with oxygen to create aluminium oxide (Al_2O_3). If a solid piece of aluminium is mechanically cut in atmosphere, the outer surface oxidises quickly to form a protective barrier, which keeps the aluminium under the oxidised surface pure.

	Warning!
	Aluminium Oxide (AlOx) coatings must not be exposed to water. Follow all recommended procedures in this manual.

Fully reacted aluminium oxide (Al_2O_3) is fairly inert, but partially reacted aluminium oxide (AlOx) can cause issues. Aluminium oxide will chemically react if it comes into contact with water, creating hydrogen gas. This reaction is exothermic, and in the normal atmosphere, the gas may self-ignite.

In the chemical reaction, the following by-products are created

- Water
- Carbon monoxide
- Carbondioxide
- HC (hydrocarbon Groups)

The clean-up required is minimal although you are requested to check local regulations.

	Warning!
	CO is an odourless gas that is harmful to humans on inhalation. The exhaust system must be sealed and kept under a negative pressure to prevent the gas coming into the work place.

Formaldehyde is another by-product that is created. Formaldehyde has a pungent odour and a fan on the exhaust system should prevent collection of any concerning quantity.

Plasma Systems

  	Warning !
	Any persons who have active or passive health aids upon their person must stay away from the equipment, when it is in operation. This is because the plasma power supply can create electromagnetic fields when operated. The risk is minimised because the electrodes are located inside the vacuum chamber and is only energised when it is closed and under a vacuum condition.

Operational Safety

	Warning!
	The following safety precautions should be adhered to at all times.

The process cycle is mainly automatic. However, there are certain mandatory safety features:

- In a production environment, the machine must not be operated with unsecured or removed guards, panels, or doors
- Before removing the winding cart from the chamber, operators must ensure there are no people or obstacles in the way
- The winding cart must be in a fully open position before attempting to load or unload a roll of substrate. [See chapter 9.4 Winding Cart Linear Traverse Drive System on page 425 for details.](#)
- When withdrawn from the chamber, the substrate winding system must only be rotated in an inch mode by the drive system, except when the drum cleaner and the safety interlock device are engaged. [See chapter 4.16 Web Drive System Inch Function on page 200 for details.](#)
- Operators must never attempt to clean the machine rollers when the winding system is rotating, or when tension is applied to the substrate, without using approved cleaning tools. [See chapter 8.1 Machine Cleaning on page 349 for details.](#)
- When an alarm siren sounds, the components may move, and all personnel should stand away from the machine
- Users must follow all guidelines given within material safety data sheets (MSDS) for the products used in the construction of this equipment
- Users must follow all guidelines for consumable materials.

A safety lockout switch is clearly labelled and is located at the front of the machine. Locking out the isolator will isolate:

- The electric motor that drives the winding cart in linear directions
- The winding system power supply.

The lockout switch also places the G-Stop circuit into a safe condition.

Before the winding cart moves, the machine will automatically trigger visual and audible alarms. The winding cart linear controls (open, close, and enable push buttons) must be depressed throughout. Automatic audible or visual alarms will automatically turn on while the substrate winding system turns in the inch mode. The inch section of the drive HMI screen must be pressed throughout. Removing the finger from the screen will automatically stop the winding mechanism. An optional remote inching push-button may be supplied within the loading area of the machine to carry out the same function.

E-Stop Safety Circuit

The E-Stop circuit stops the electrical power to the rotating equipment. All users must know the location of all E-Stop buttons.

When the E-Stop button is pressed, the control circuit will switch off power to the winding system drive circuits. The rollers and rolls of substrate will slow and all other rotating equipment will be powered down.



Figure 1.5-1 The E-Stop button on the operator interface panel

The button will isolate:

- Vacuum pumps
- Process chiller
- Plasma treaters
- Winding mechanism drive system
- Aluminium wire feeders
- Evaporation power supply
- Winding cart traverse movement.

Resetting after an E-Stop Condition

	Warning ! Ensure that the machine is in a safe condition.
---	---

When the machine needs to be reset, the E-Stop Reset button will flash on the operator panel. Press the button to reset the E-stop circuit and enable the electrical circuits.

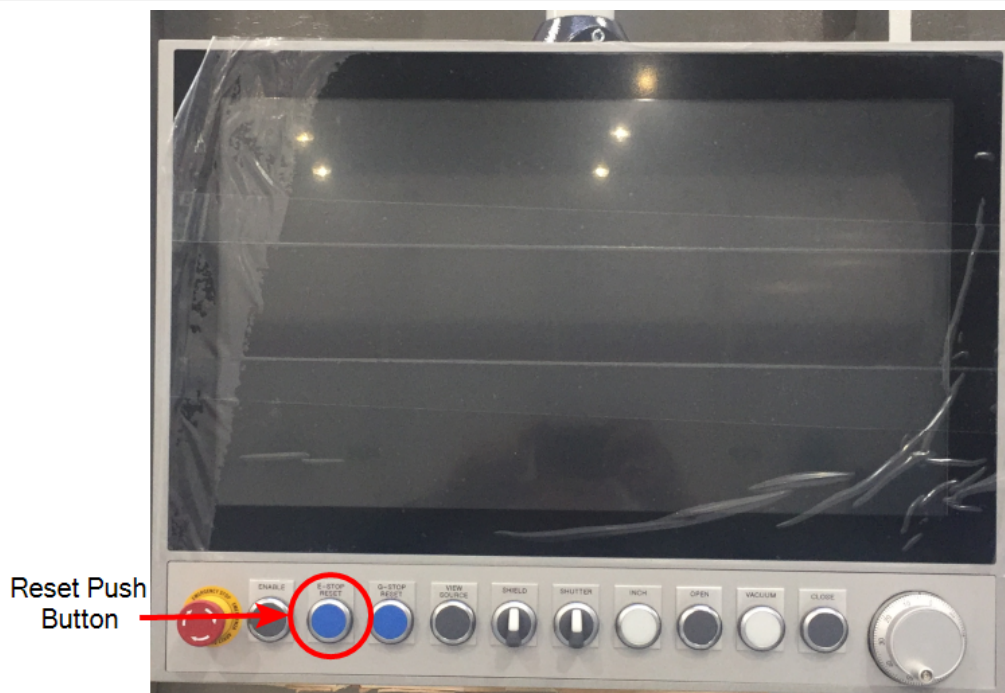


Figure 1.5-2 The E-Stop Reset button on the operator interface panel

Resetting on Machine Start Up

The operation is the same on machine start up.

G-Stop Safety Circuit

The G-Stop circuit is triggered by rotating the traverse movement isolator on the E06_0 panel to the off position. The circuit de-energises the following:

- winding cart traverse movement, into and out of the chamber
- coating shield rotational movement
- high vacuum valve VV7 movement
- coating shutter open and close movement
- evaporation source area viewing flap movement
- winding mechanism system movement
- web threader movement

Resetting

	Warning !
	Ensure that the machine is in a safe condition.

When the machine needs to be reset, the G-Stop Reset button will flash on the operator panel. Press the button to reset the G-stop circuit and enable the electrical circuits.

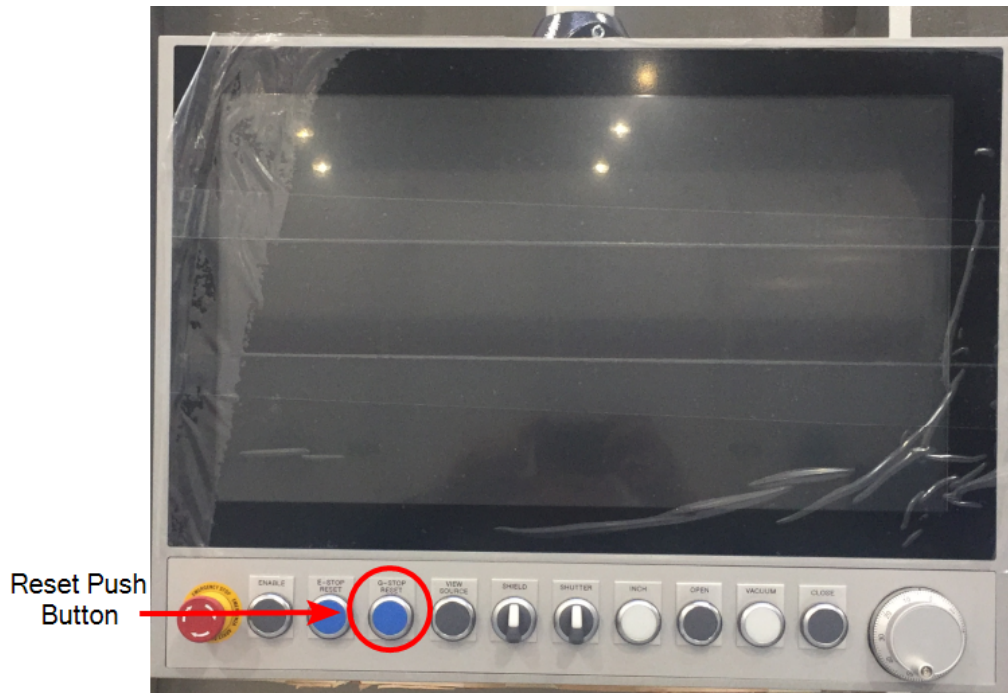


Figure 1.5-3 The G-Stop Reset button on the operator interface panel

Resetting on Machine Start Up

The RESET push button operation is the same on machine start up.

Locking Out the Movement Isolator

	Warning ! No person should enter the chamber unless they have locked out the isolator into a safe condition and applied a suitable padlock.
---	---

To lockout the movement lockout isolator, turn it through 90 degrees and place a padlock on the switch.

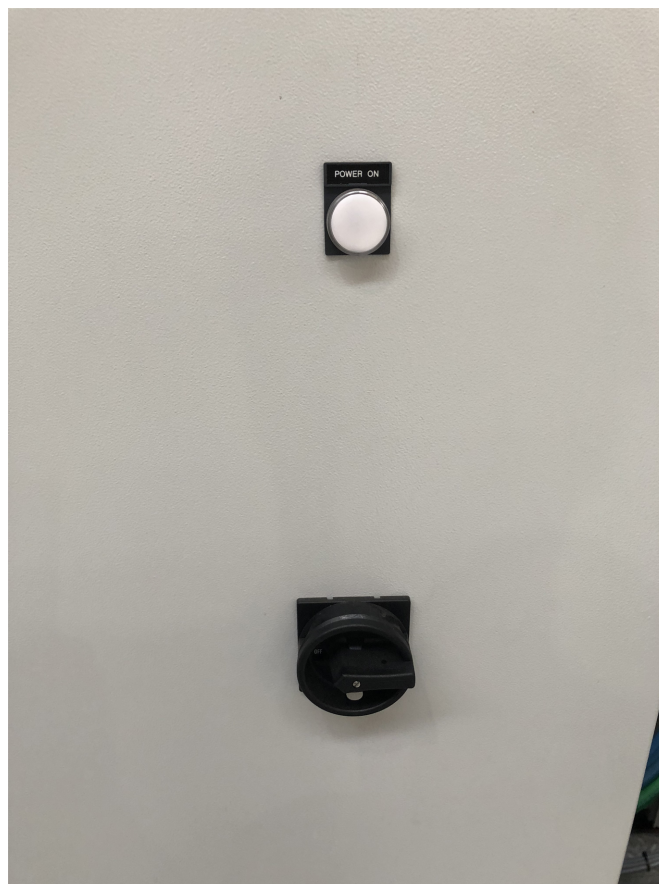


Figure 1.5-4 Locking Out Of Movement Isolator

Audible Siren and Visual Warning Beacons

Audible alarms and visual warning beacons will operate in various conditions:

Visual Beacon Condition	Audible Alarm Condition	Meaning
Off	Intermittent: two seconds on, two seconds off	Lock out isolator has not been rotated to safe condition.
Flashing	On	Stand away from the machine. Movement is imminent and one of the following conditions will occur: - tension applied to the substrate - movement of winding cart - inching of the substrate winding system - movement of layarm roller assemblies.

Table 1.5-2 Warning Siren and Beacons

Legal Notice

	Important!
	Only fully accredited personnel should carry out inter-connection of wiring on any electrical equipment. All electrical equipment should be switched off at the mains isolator switch before any adjustment or connections are made to the control panels or electrical components.

Under Section 6(1) and 6(4) of the UK Health and Safety at Work Act, we make available certain information on the product we supply. The following information must be brought to the attention of all personnel involved with this equipment as required by the Act:

- all users should be aware of the hazards present and act accordingly
- the equipment is electrically powered at the local main supply voltage; the machine must be earthed and all local electrical supply regulations must be complied with
- all protective guards, electrical or mechanical interlocks, and other safety devices that were fitted when the machine was delivered must be kept in place throughout its operational life cycle
- safety devices must be inspected for effective operation at regular intervals by competent maintenance personnel
- equipment must be completely isolated from the electrical supply before any terminal covers over live electrical components are removed
- equipment should not be tampered with, modified, or used in a non-standard manner without first seeking advice from Bobst Manchester
- all hoses and ancillary devices that are connected to the equipment should be of suitable quality and rating to withstand the operating temperature and internal pressure specified in data sheets provided
- flexible hoses connected by the user should be installed to conform to the recommendations of the hose manufacturers, especially with regard to minimum bending radii, and avoidance of wear by friction of moving hose or adjacent parts
- all pipe work, valves, flexible hoses, etc., should be regularly inspected by qualified personnel
- if damage or leakage occurs, the equipment should be taken out of service until rectified
- warning plates must not be defaced, removed, or painted.

Safety Lockout Procedure

	Warning!
	Never remove another person's lockout padlock or tag. Never assume that a machine is locked out.

Users should be familiar with their company's lockout procedure. The following is an example. Bobst Manchester does not accept any responsibility for loss, damage, or injury if this procedure is inaccurate:

- remove all rolls of substrate from the winding system
- lock out the main power supply isolator with your own padlock
- attach a tag on the lock with your name and the words 'Danger - Personnel Working'
- block rotating elements to prevent accidental rotation
- block components that may lower under their own weight
- shut off all sources of power and flow of material and services, including
 - electrical power supplies
 - pneumatic air supplies
 - process gas supply systems.

Pneumatic Air Supply Isolation Valve

A single air supply isolation valve is located on the vacuum pumping system. It is normally positioned on a framework supporting a pump. The valve is clearly marked with a warning label.

Before isolating the air supply, ensure that any equipment on the machine that is supported by a pneumatic device is in a safe condition.

Diffusion Pump Oil

	Warning!
	All diffusion pumps use silicone-based oils. No mineral-based oils should be used.

Potential Health Hazards - General

Extremely fine particles are deposited on the chamber walls and other surfaces of the vacuum system. Be aware of the following health hazards:

- some substances can cause lung damage; wear a protective respirator mask with fine filter approved by a relevant safety organisation to avoid inhaling fine particles or powder
- avoid inhalation of toxic substances; refer to material data sheets
- when opening the chamber after a cycle, allow time for the coating surface to oxidize; failure to do so may cause flash fires when the material is exposed to oxygen
- breakage of reactive condensates may be hazardous, even when precautions are observed; refer to material safety sheets and wear fireproof clothing.

Handling and Use of Oxygen Process Gases and Distribution Systems

Oxygen gas lines and regulators are subject to specific safety guidelines. Bobst Manchester follows industry practices in the design and installation of these lines, and customers should do likewise. Refer to the guidance section in this manual. Failure to follow these guidelines can result in accidents.

Noise Reduction

Hearing protection must be worn in environments with noise levels exceeding 82 decibels, measured as a weighted average over an 8-hour period. The machine is designed to operate within this range to avoid the need for hearing protection. However, reflective surfaces around the machine may increase noise levels.

Various components generate noise, including:

- Drive motor blowers, which create noise by forcing air through at high speed
- Chiller unit compressor and cryogenerator, which should have cover fitted at all times except when maintenance is being carried out
- Belt drives, which can generate noise due to improper tension; use the guidance in this manual to resolve this
- Belt guards, which can create a squealing noise if they touch a rotating member
- Pumps, which will not generate noise above the allowed level unless there is a fault, such as a slipping drive belt
- Vents, which are designed with filters to reduce noise.

Auxiliary equipment such as compressors and water supply pumps can also produce noise and should be housed within an acoustic enclosure.

Do not obstruct grilles or use restrictors in an effort to reduce noise. If noise is caused due to a suspected fault, this should be immediately investigated.

Risk Analysis

	Important! This equipment is constantly being reviewed and improved. For more information on equipment, safe usage, or installation, contact BOBST.
---	--

	Warning! Do not bypass or remove safety systems.
---	---

BOBST carries out a risk analysis of potential hazards. If it is not possible to completely eliminate a hazard, the risk is mitigated. Warning signs, and instructions in this manual, are designed to make users aware.

FORWARD NOTES

Following the operating and preventative maintenance guidelines in this manual will ensure long and satisfactory operation. When carrying out maintenance work, ensure that safety precautions are observed and that electrical power, hydraulic, air and water are isolated. Users operating and maintaining this machine should initially read this manual and receive appropriate training.

	Note!
	Bobst Manchester cannot accept responsibility for instructions supplied by manufacturers of equipment that has been incorporated into the machine.

Definition of Personnel

Bobst Manchester Ltd recommends that persons operating and maintaining this machine are grouped into three categories:

- Operating personnel
- Production supervisors
- Maintenance and service personnel.

Descriptions of each group follow in the next section on this page. Users in a group are required to have the knowledge and skills of the lower groups.

The rights of personnel to access the various levels of the machine control are defined by assigning login passwords.

Access Rights	Personnel		
	Operator	Production Supervisor	Maintenance Personnel
Vacuum System – Automatic Control	✓	✓	✓
Vacuum System – Manual Control	✗	✓	✓
Recipe Setup – Normal Access	✓	✓	✓
Recipe Setup - File Deletion	✗	✓	✗
Maintenance Timers/Configuration	✗	✗	✓
Deposition System Calibration	✗	✓	✓
Pump down Reference Curve Save	✗	✓	✓
Shutdown WinCC - Interface	✗	✓	✓
Calibrate Monitor Systems	✗	✓	✓
Change Maintenance Timers	✗	✗	✓
Sign off Maintenance Items	✗	✗	✓
Change Configuration	✗	✗	✓
Water Isolation Valve Operation	✗	✓	✓

Table 1.6-1 Levels of Access to Control Systems

Definition of Operating Personnel

Operating personnel are permitted to operate the machine in normal operation (automatic coating). This comprises the following activities:

- Carrying out coating process via the Human Machine Interface (HMI) and other parts of the machine
- Loading and unloading the winding system with substrate
- Cleaning the machine as described in this manual
- Replacing the pressurized process gas bottles

Operating personnel must be trained to operate the machine by an experienced supervisor that has previously been trained by an employee of BOBST.

Definition of Production Supervisors

Production supervisors are permitted to operate the machine in normal operation in the same way as operating personnel. In addition, the supervisor may:

- Access all the normal functions of the operator station
- Create, store and delete process recipes
- Check the calibration of the machine systems.

Production supervisors must have the same operational knowledge as operating personnel. They must have been trained by an employee of BOBST.

Definition of Maintenance Personnel

	Warning! Reduced level of safety. Improper operation may cause serious injuries and equipment damage.
--	---

Maintenance or service personnel have a limited knowledge of the machine operation. They may use the machine in normal conditions, perform maintenance work, and perform service work. For this purpose, the machine may be operated in certain manual modes.

Maintenance of the machine means:

- Switching the system on and off
- Carrying out preventative maintenance as described in this manual.

Service of the machine means:

- Shutting down and dismantling the machine
- Accessing system components
- Repairing mechanical and electrical components.

Only trained employees may maintain and service the machine. Maintenance personnel that work on electrical components should complete suitable training as electrical technicians or similar.

Equipment or Parts Return

To return a part to Bobst Manchester:

1. Contact the service department at services.manchester@bobst.com to obtain a returns authorisation number (RMA)
2. Pack the return, enclosing a note stating the RMA
3. Label the package with the sender's address
4. Return the part to Bobst Manchester Ltd., Pennine Business Park, Heywood, Lancashire, OL10 2TL, UK.

SECTION 1.0 INTRODUCTION & GENERAL DESCRIPTION

This section of the manual introduces processes and major machine components.

- [1.1 Physical Vaporisation Process on page 30](#)
- [1.2 Practical Evaporation Process Description on page 31](#)
- [1.3 Identifying Machine Sections on page 32](#)
- [1.4 Substrate Winding System on page 34](#)
- [1.5 Vacuum Pumping System on page 36](#)
- [1.6 Aluminium Evaporation Source on page 37](#)
- [1.7 Machine Control Systems on page 38](#)

Please refer to [Section 9.0 Auxiliary equipment on page 369](#) for information about the auxiliary components of the machine.

1.1 Physical Vaporisation Process

The machine produces a substrate that can be used for stamping foils, security holograms, decorative packaging, and barrier packaging using a roll to roll (R2R) process.

Solid aluminium wire is fed onto ceramic evaporators (boats) and heated to approximately 1500°C. The aluminium melts into a liquid. In the high vacuum environment inside the machine, the aluminium turns into a vapour. This evaporation process is called physical vapour deposition (PVD).

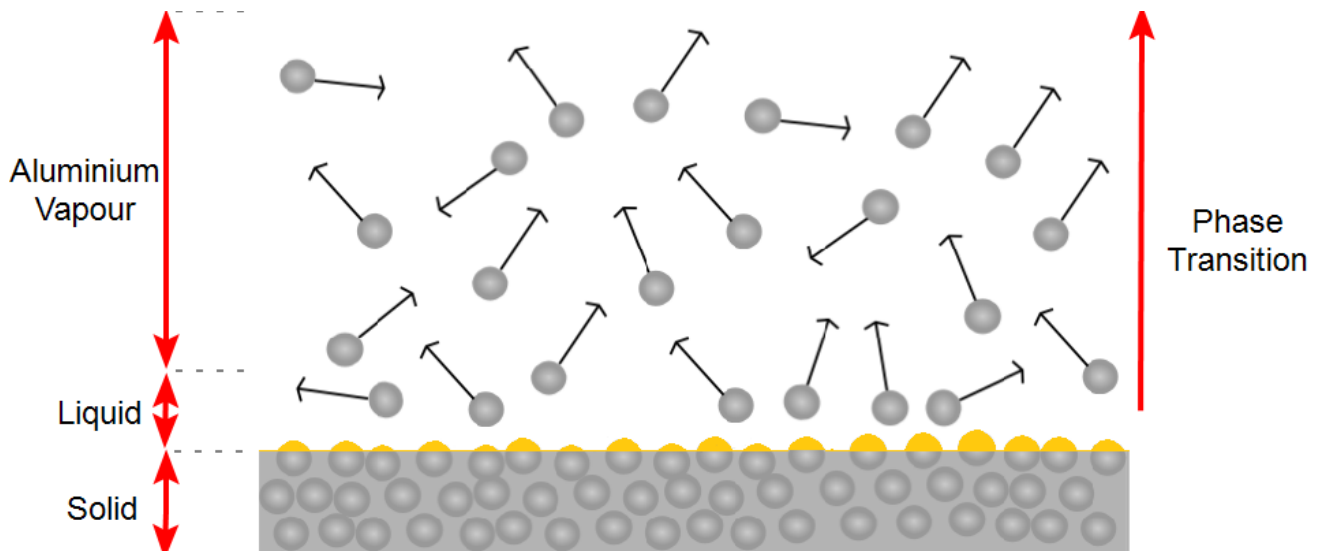


Figure 1.1-1 Indication of phase changes

The air above the liquid aluminium has a reduced molecular density due to the vacuum in the chamber. This forms a cloud of aluminium vapour that rises with heat. The substrate is passed over the vapour cloud and a thin film of aluminium adheres to the surface.

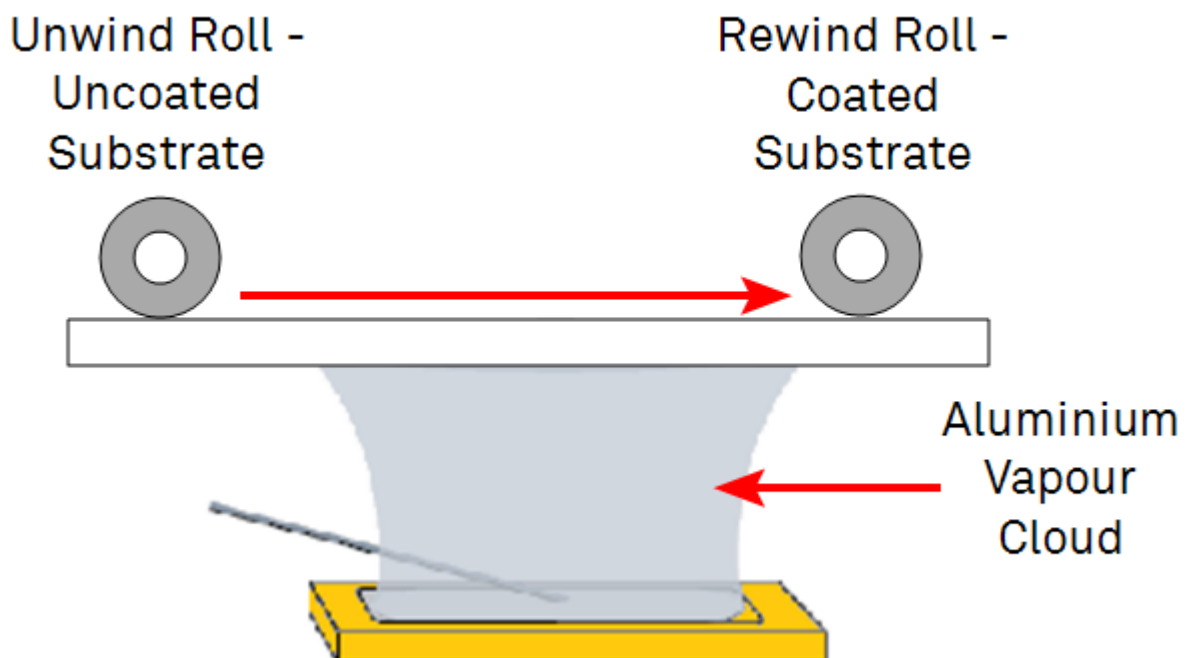


Figure 1.1-2 Drawing showing basic concept of PVD process

Under normal process conditions, variables are modulated to create an even, uniform coating across the width of the substrate. The machine uses a cooled process drum to stop the substrate from being damaged by heat.

1.2 Practical Evaporation Process Description

In the evaporation process, heated ceramic evaporators process a metal in a high vacuum environment (5×10^{-4} mbar). The metal is usually aluminium. The process takes place in a vacuum chamber.

The substrate moves past the coating area. This allows the aluminium vapour to condense on the moving web of plastic or paper.

The substrate moves past the evaporators at high speed. It receives radiated heat from the evaporators, and latent heat from the condensing aluminium.

The substrate must be properly tensioned between rollers as it passes through the coating area. The winding system transfers the substrate from the unwind roll, over tension measuring rollers, over spreader rollers, through the coating area, and finally to the rewind roll. This takes place without stretching or damage.

This is a batch process. One machine can process a roll of substrate in approximately 60 minutes, depending upon the size of the roll and the coating deposition/ speed.

Each evaporator has a limited capacity, so the machine has several. The evaporators are positioned at a 90 degree angle to the substrate movement.

Evaporators are evenly spaced apart and extend across the width of the machine to allow even coverage.

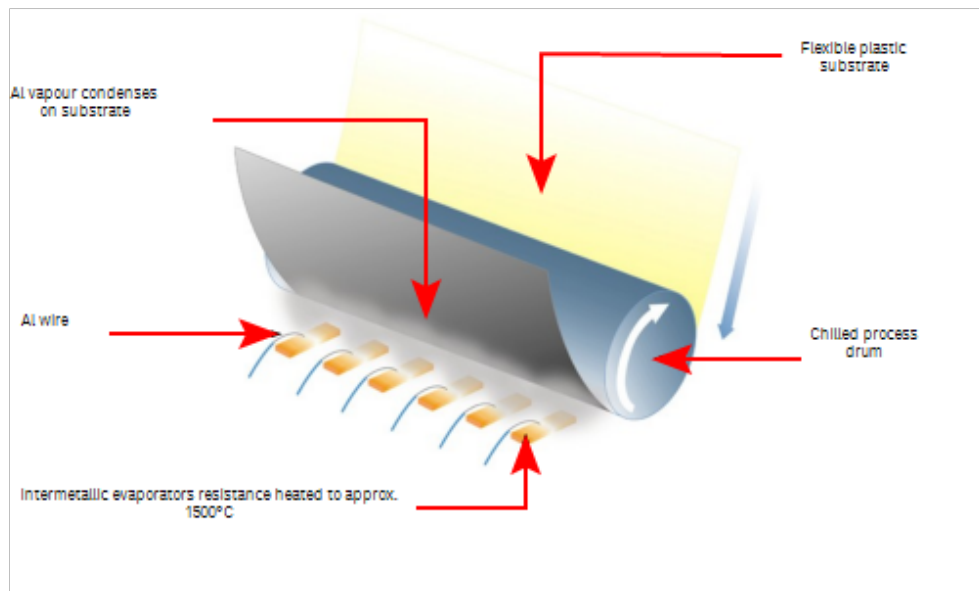


Figure 1.2-1 Physical vapour evaporation deposition process schematic drawing

1.3 Identifying Machine Sections

The evaporation process takes place in a vacuum tank, or chamber. The chamber is made from mild steel in a reinforced rectangular shape. This ensures that atmospheric pressure is evenly distributed on the outside, and minimises mechanical distortion.

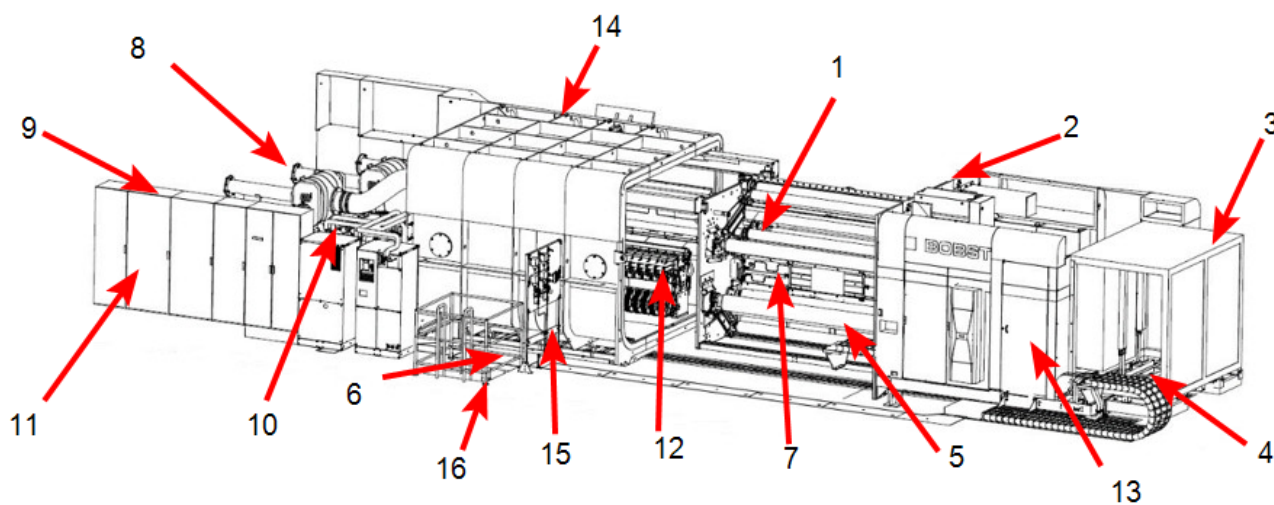


Figure 1.3-1 Rear view of the K5 Expert with the cart withdrawn from the chamber

Reference	Description
1	Winding system rewind roll position
2	Winding cart faceplate
3	Process chiller unit
4	Cable coiler location
5	Winding system unwind roll position
6	Diffusion vacuum pumps – under chamber
7	Location of pre-plasma treater behind unwind roll
8	Vacuum pumping system
9	Electrical panel E03_0
10	Water supply isolation valves
11	Main Electrical isolator
12	Evaporation enclosure
13	Rear HMI interface panel
14	Chamber
15	Process gas distribution
16	Access to source transformers

Table 1.3-1 List of components on machine - rear view

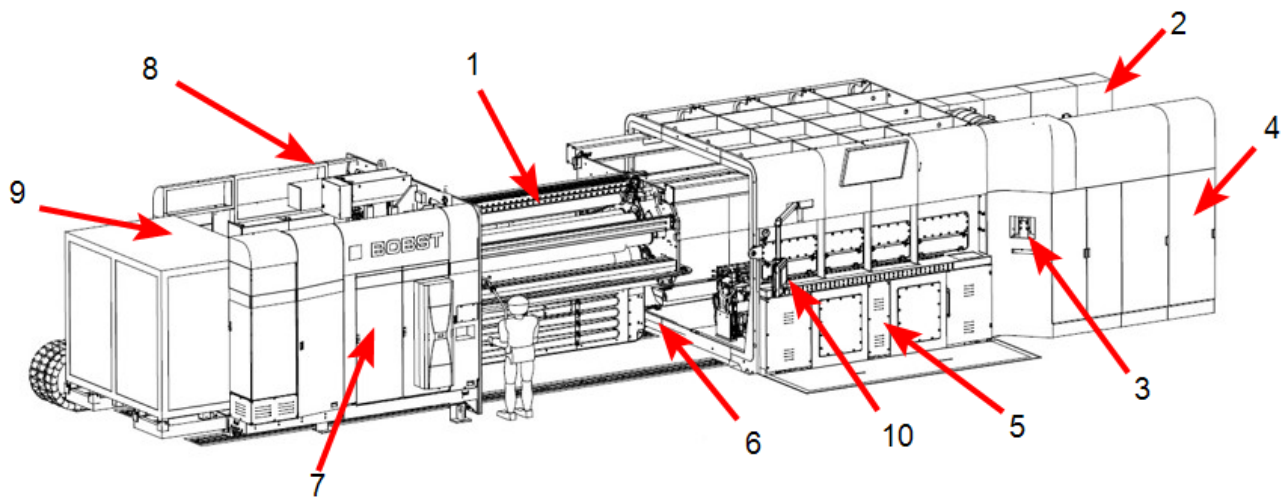


Figure 1.3-2 K5 Expert front view

Reference	Description
1	Location of inline deposition monitor
2	Location of vacuum pumping system
3	Brookes cryogenerator unit
4	Electrical panel E03_2
5	Access cover to VV7 valve assembly
6	Separation wall between vacuum zones
7	Electrical panel E06-0
8	Unwind and rewind drive motor location
9	Plasma treater power supply
10	Operator Panel E05_0

Table 1.3-2 List of components on machine - front view

1.4 Substrate Winding System

The substrate is wound in rolls. The rolls are loaded onto shafts that are then loaded onto the unwind chucks. The substrate is threaded through the winding system to the rewind roll.

Winding System Components

The substrate must be properly tensioned between rollers as it passes through the physical vapour deposition zone. The winding system transfers the substrate from the unwind roll, over tension measuring rollers, over spreader rollers, through the PVD zone, and finally to the rewind roll. This takes place without stretching or damage.

The winding mechanism includes the following components:

- unwind roll
- spreader rollers
- load cell rollers
- idler rollers
- water cooled rollers
- tensioning rollers
- rewind roll
- layon rollers

The layarm rollers move close to winders and support the web.

Drum tensioning rollers (DTRs) apply tension to the substrate to hold it tightly on the process drum surface.

Some of the rollers are driven by an AC drive system, including:

- unwind roll
- cooling drum
- cooled post drum roller
- cork covered draw roller
- rewind roll.

The electrical drive control system used is composed of standard hardware components and uses industry standard software to provide accurate winding conditions.

Other rollers within the mechanism are designed from lightweight materials and are mounted on low friction bearings.

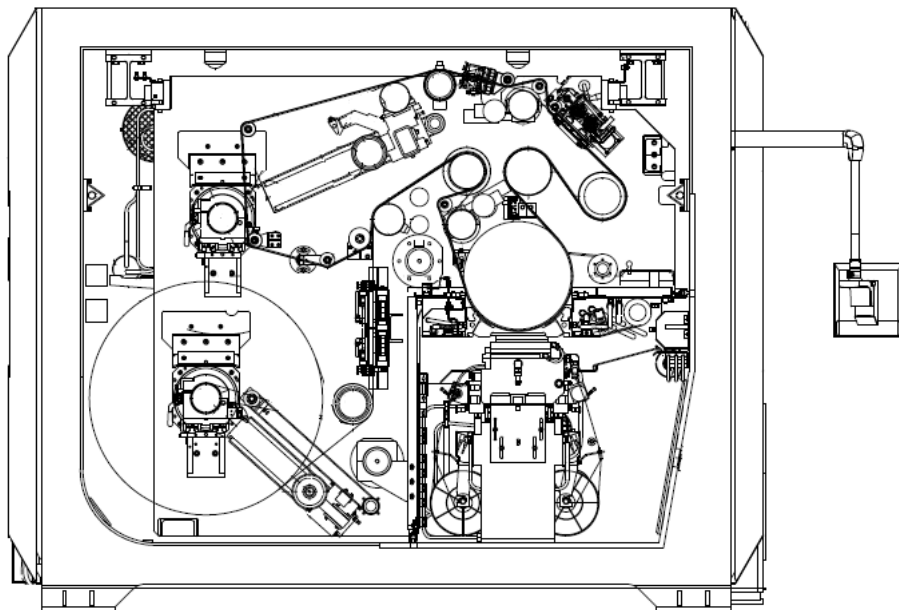


Figure 1.4-1 Cross-section of the K5 Expert winding mechanism

Winding Cart

The winding system is located inside the winding cart.

The winding cart can be driven into and out of the vacuum chamber. It is withdrawn from the vacuum chamber to allow for rolls of substrate to be placed in the machine. It is driven into the chamber to allow the vacuum to be created and start the vapourisation process.

The winding cart has two different types of inter-zone seals:

1. rigid, low-conductance concave aluminium seals are located close to the drum where the substrate enters and exits the evaporation zone
2. flexible inflatable sealing strips inside the chamber provide flexible zone seals with the winding cart.

The faceplate and cart are used to support ancillary equipment for the machine:

- web transport mechanism
- process chiller unit(s)
- web transport drive system; electrical panel and drive motors
- plasma treater power supply components
- cryo-generators
- control systems

The documentation refers to a faceplate side and an outboard side, corresponding to two large plates that the winding rollers are located into.

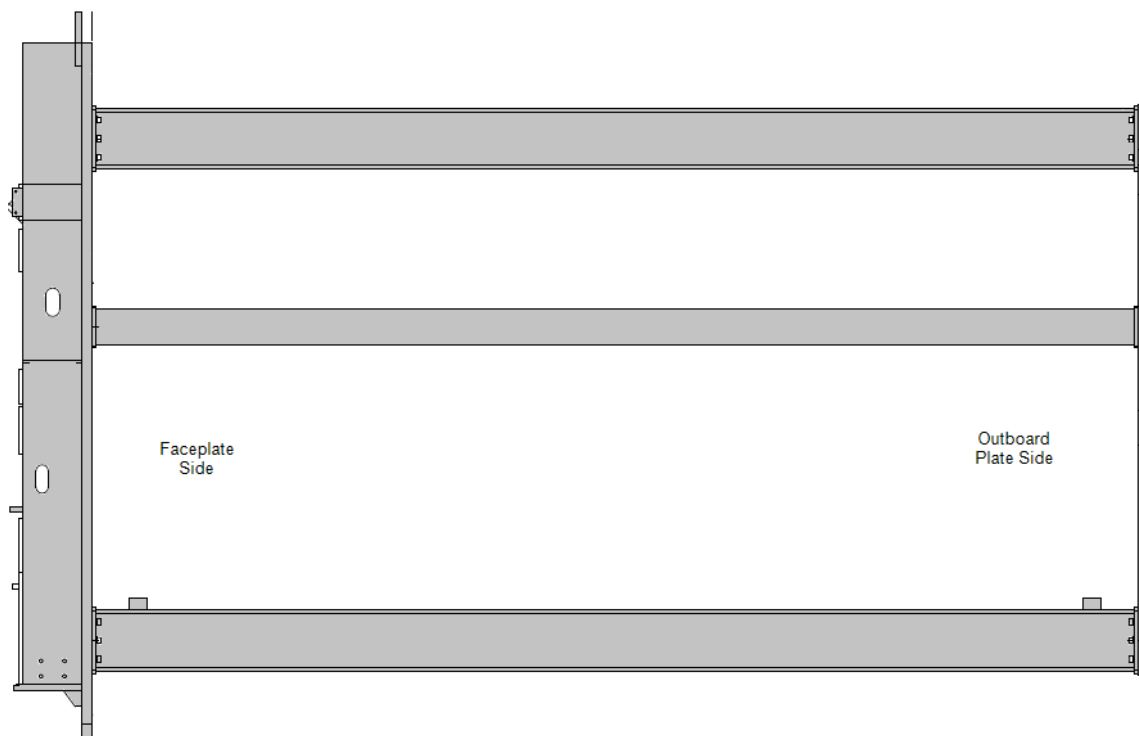


Figure 1.4-2 K5 Expert winding mechanism framework

1.5 Vacuum Pumping System

Different types of pumps are used to evacuate the chamber and create the vacuum environment for the coating process. See [2.5 Types of Pumps Incorporated on Machines on page 50](#) for a description of these pumps.

The vacuum chamber is divided into two zones by various sealing methods within the chamber. This allows for a number of benefits:

- low power consumption due to smaller pumping group requirements
- better vacuum pressure levels in the evaporation zone
- improved cleanliness of the winding zone.

The HMI screen provides an overview of the vacuum system status.

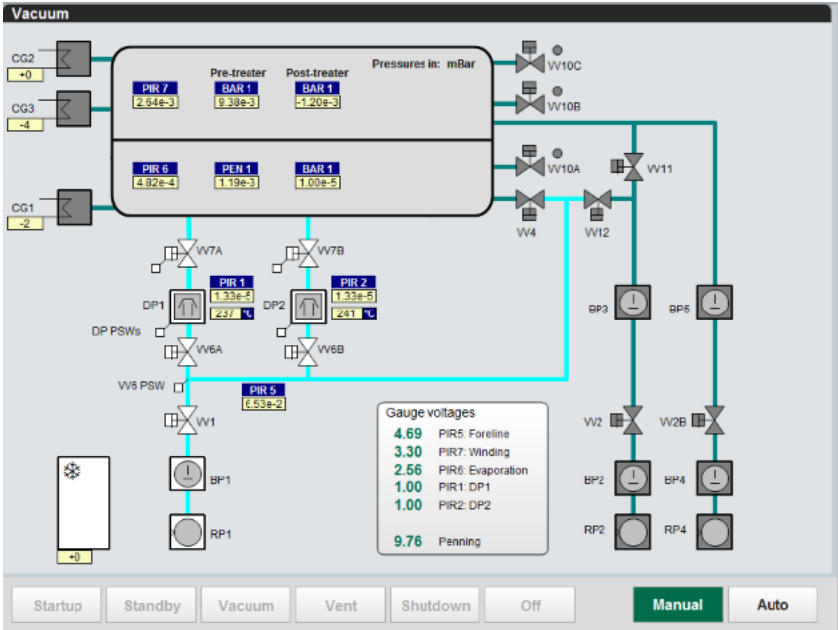


Figure 1.5-1 Vacuum pumping system HMI screen

[See chapter 2.2 Vacuum Pumping System Mimic Display Screen on page 43 for details.](#)